Information Regarding Insulin Storage and Switching Between Products in an Emergency

https://www.fda.gov/drugs/emergency-preparedness-drugs/information-regarding-insulin-storage-and-switching-between-products-emergency

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Library safety context: General emergency reference, not individualized treatment advice. Insulin storage, concentrations, devices and switching rules are product-specific. Follow current product labeling and your clinician or pharmacist; do not infer substitutions from a class name alone. Read the complete source, including warnings. DKA is an emergency requiring immediate medical attention.

Insulin for Injection | Insulin Switching | Insulin Mixes | Insulin Pumps

This guide explains how to store and use insulin during an emergency, when your usual supply may not be available. Always follow your doctor's instructions when possible.

Insulin Storage and Effectiveness

During a disaster, you may receive insulin from a different manufacturer than what you normally use. You may also not have access to a refrigerator. Here is what you need to know about storing insulin safely.

Insulin for Injection

According to the product labels from all three U.S. insulin manufacturers, it is recommended that insulin be stored in a refrigerator at approximately 36°F to 46°F. Unopened and stored in this manner, these products maintain effectiveness until the expiration date on the package.

Insulin products contained in vials or cartridges supplied by the manufacturers (opened or unopened) may be left unrefrigerated at a temperature between 59°F and 86°F for up to 28 days and continue to work. However, an insulin product that has been diluted or removed from the manufacturer’s original vial should be thrown away within two weeks.

Extreme Temperatures

Extreme temperatures make insulin less effective. The longer insulin is exposed to extreme heat or cold, the less effective it becomes. Over time, this can make it harder to control your blood sugar. Under emergency conditions, you might still need to use insulin that has been stored above 86°F.

You should try to keep insulin as cool as possible. If you are using ice, avoid freezing the insulin. Do not use insulin that has been frozen. Keep insulin away from direct heat and out of direct sunlight.

Once you have access to properly stored insulin, throw away any vials that were exposed to extreme temperatures and replace them as soon as possible. If patients or healthcare providers have specific questions about the suitability of their insulin, they may call the manufacturer at the following numbers:

Manufacturer (Product(s)): Customer Service Number(s)

Manufacturer | Products | Customer Service

Biocon Biologics | Semglee, Kirsty | 1-833-986-1468

Meitheal Pharmaceuticals | Garzulys | 1-844-824-8426

Lanexa Biologics, LLC | Langlara | 1-800-610-5709

Lilly | Humalog, Humulin, Lyumjev, Basaglar, Rezvoglar | 1-800-545-5979

MannKind Corporation | Afrezza | 1-844-323-7399

Novo Nordisk | NovoLog, Fiasp, Novolin, Tresiba, Awiqli | 1-800-727-6500

Sanofi | Lantus, Toujeo, Apidra, Admelog, Merilog | 1-800-633-1610

Note : Manufacturer contact numbers are subject to change. Patients and healthcare providers should verify current contact information using the product's package insert or by visiting the manufacturer's official website.

Special Storage: Once-Weekly Insulin (Awiqli)

Awiqli (insulin icodec) is only available as a prefilled pen. It has a longer unrefrigerated storage window than other insulin products. Unopened Awiqli FlexTouch pens stored in a refrigerator (36°F to 46°F) may be kept until the expiration date. Once removed from the refrigerator, or when in use, Awiqli pens may be stored at room temperature below 86°F for up to 12 weeks. The total combined time at room temperature, whether opened or unopened, must not exceed 12 weeks. Do not freeze Awiqli; do not use if it has been frozen.

Note : The in-use pen may also be stored in a refrigerator (36°F to 46°F) for up to 12 weeks — the 12-week limit applies regardless of whether storage is at room temperature or refrigerated.

Special Storage: Insulin Pumps

Insulin in the pump’s infusion set (such as the reservoir, tubing, catheters) should be thrown away after 48 hours. Unless your pump device’s instructions are different. For example, NovoLog, Kirsty, and Humalog U-100 specify up to 6–7 days in the reservoir; Fiasp PumpCart specifies 4 days. Always follow the specific product's prescribing information and your pump device's instructions for use.

If the pump insulin is exposed to temperatures above 98.6°F, throw it away.

Switching Insulin

Always switch insulin under a doctor’s care, with close monitoring of your blood sugar. If you cannot reach a doctor during an emergency, follow the guidance below — but monitor your blood sugar closely and see a doctor as soon as you can.

Short-acting (regular insulin) and Rapid-acting Insulins

In an emergency, you can switch one brand of regular insulin (such as Humulin R or Novolin R) for another brand of regular insulin, or for a rapid-acting insulin (such as Apidra, Fiasp, Humalog, or NovoLog), and vice versa. Use the same number of units.

Inject regular insulins approximately 30 minutes before the start of each meal. Rapid-acting insulins begin working more rapidly than regular insulin and are to be injected no more than 15 minutes before the start of each meal to avoid dangerously low blood sugar levels.

Intermediate and Long-acting Insulins

In an emergency, you can switch one brand of intermediate-acting insulin (such as, Humulin N, Novolin N) for another intermediate-acting insulin on a unit-per-unit basis. You can also switch intermediate-acting insulins for long-acting insulins (e.g., Lantus, Toujeo, Tresiba) on a total unit-per-day basis, or vice versa.

Important :

• If you switch from a once-daily long-acting insulin (e.g., Lantus, Toujeo, Tresiba) to an intermediate-acting insulin, keep your total daily dose the same but split it into two equal injections. Give half in the morning with breakfast and half in the evening with dinner to avoid a dangerously low blood sugar.

• If you switch from an intermediate-acting to a once-daily long-acting insulin, add up the total amount of intermediate-acting insulin units for one day, and give it as a single long-acting insulin dose once a day.

Close monitoring of blood sugar and adjustment of your insulin dose may be needed in the transition.

Once-weekly Long-acting Insulin

If you use the once-weekly long-acting insulin, Awiqli (insulin icodec) and need to switch to a once-daily long-acting or intermediate-acting insulin in an emergency, base your dose on your total weekly dose.

When switching from a once-daily long-acting insulin to insulin icodec, give the first insulin icodec dose on the day after your last daily long-acting insulin dose. For Week 1, multiply your previous total daily dose by 7, then by 1.5, and round to the nearest 10 units.

Important: Because insulin icodec stays active for about one week, its blood sugar-lowering effect continues for several days after the last dose. If you switch from Awiqli to a basal (background) long-acting insulin, switching should be done under close medical supervision with increased blood sugar monitoring to avoid low blood sugar (hypoglycemia).

Note : Insulin icodec must not be used in an insulin pump, mixed with other insulins, or drawn from the pen into a syringe.

Insulin Mixes

Switching between types of insulin should be done with a doctor’s guidance and close blood sugar monitoring.

What is an Insulin Mix?

Insulin mixes combine two types of insulin, intermediate- and short/rapid-acting, in one dose. The first number shows how much intermediate insulin is in the mix and the second number shows how much short/rapid-acting insulin. For example, each dose of a 70/30 mix contains 70% intermediate-acting insulin and 30% short/rapid-acting insulin.

Switching from Your Insulin Mix: Substitution or Replacement

If you use a pre-mixed insulin (e.g., Humulin 70/30, Humalog Mix 75/25, Novolin 70/30, NovoLog Mix 70/30), here are your options in an emergency:

• In an emergency, you can switch one insulin mix for another on a unit-per-unit basis.

• Insulin mixes with rapid-acting insulin analog (e.g., Humalog Mix, Novolog Mix) should be injected closer (within 15 minutes) to the start of the meal compared to mixes containing regular insulin (e.g., Humulin 70/30).

• If an insulin mix is not available, patients should follow this two-step process:

• Step one: replace an intermediate-acting part of the mix (about 70% of the total units for each dose) with an intermediate-acting or a long-acting insulin on a unit-per-unit basis.

• With an intermediate-acting insulin :

• Give 70% of total units for each dose

• With a long-acting insulin :

• Add up the total insulin units given in one day

• Give 70% as one daily dose

• Step two: if regular or rapid-acting insulins are also available, use them before major meals. The dose should be about 30% of your total of pre-mixed insulin for that meal, in addition to the intermediate- or long-acting.

Note : Inject longer and shorter acting insulins separately unless your doctor tells you otherwise.

Insulin Pumps

If your pump supplies are not available or your pump is not working properly, you may need to switch to injected insulin. This should be done with a doctor's guidance and close blood sugar monitoring whenever possible.

Using a Different Insulin in Your Insulin Pump

I n an emergency, you may be able to swap one rapid-acting insulin (such as Apidra, Fiasp, Humalog, Novolog) for another on a unit-per-unit basis . Check your pump’s instructions to make sure the available insulins are compatible with your devices.

Note : Not all rapid-acting insulins are approved for pump use. Merilog (insulin aspart-szjj) is not currently labeled for insulin pump use. Humalog U-200 and Lyumjev U-200 must NOT be used in insulin pumps. Always check the specific insulin product labeling before use in a pump.

Switching from an Insulin Pump to Injected Insulin

If you must switch from your pump to injected insulin, use the following guidance:

• Replace your pump’s total 24-hour basal (background) dose with insulin on a unit-per-unit basis.

• For intermediate-acting insulin, split the 24-hour basal (background) dose into two equal injections per day. For example, if your 24-hour basal (background) dose is 24 units, inject 12 units twice daily.

• For long-acting insulin, give the full 24-hour basal (background) dose as one injection per day.

• If regular or rapid-acting insulin is also available, use it with each meal. Match your pump’s bolus (mealtime) dose on a unit-per-unit basis.

• For example, if your pump gives a bolus (mealtime) of 5 units to cover breakfast, inject 5 units of regular or rapid-acting insulin to cover that meal

Resources

• FDA-Approved Insulin Products (PDF - 378 KB)

Source links (require internet)

FDA-Approved Insulin Products: https://www.fda.gov/media/195052/download?attachment